

A CNN never looks at the **whole image**

FINDING A VERTICAL EDGE



The SAME small window slides everywhere — that is why CNNs need so little data

- 1 It slides a window a few pixels wide, looking for **one pattern**.
- 2 On flat colour: nothing. Straddling a boundary: **edge found**.
- 3 Stack many windows and you detect many patterns at once.

REMEMBER THIS

The SAME small window slides everywhere. That is why CNNs need so little data.